

Practical Demo: Corporate-Bond Spread Risk and Scenario Analysis

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This reading copy shows the calculations' results and figures. The HTML demo contains the full code in folded blocks and interactive scenario controls.

The desk question

Which positions dominate credit risk, and how quickly can spread widening consume their carry?

Learning Objectives

After completing this demo you should be able to decompose corporate-bond risk into Treasury-rate, spread, default, and liquidity components; calculate spread duration and duration times spread (DTS); compare spread carry with widening risk; and evaluate mild and severe portfolio scenarios.

1 Inputs and Assumptions

The decision: a credit manager is offered more spread by moving into a lower-rated bond. Does the extra carry justify the loss if spreads widen? Follow the chain **position** → **dollar sensitivity** → **carry cushion** → **stress loss**. For each step, predict the sign before opening the folded calculation.

All values are **illustrative, market-like inputs**, not current quotes. The portfolio is intentionally small enough to audit by hand. Prices are near par; annual carry is approximated by spread times market value; spread duration is treated as constant; Treasury and spread P&L use first-order duration; PDs are physical one-year probabilities; defaults and mark-to-market spread changes are reported separately to avoid double counting.

Workflow stage	In this case
Observed or supplied	Market value, benchmark spread, and security terms
Estimated	Spread duration, PD, recovery, and liquidity reserve
Diagnosed	DTS concentration and carry breakeven
Used for	Mild and severe portfolio stress decisions

Bond	A utility 2029	BBB bank 2031	BBB industrial 2034
Market value (\$mm)	12	10	8
Treasury duration	4.2	5.6	7.1
Spread duration	4	5.1	6.3

Bond	A utility 2029	BBB bank 2031	BBB industrial 2034
Spread (bp)	78	142	185
One-year PD	0.0015	0.004	0.007
Recovery rate	0.4	0.4	0.35
Liquidity reserve (bp/year)	8	15	22
Treasury DV01 (\$/bp)	5,040	5,600	5,680
Spread DV01 (\$/bp)	4,800	5,100	5,040
DTS	0.0312	0.0724	0.1166
Contribution to DTS	0.0107	0.0207	0.0266

Bond	BB telecom 2028
Market value (\$mm)	5
Treasury duration	3.4
Spread duration	3
Spread (bp)	410
One-year PD	0.025
Recovery rate	0.3
Liquidity reserve (bp/year)	45
Treasury DV01 (\$/bp)	1,700
Spread DV01 (\$/bp)	1,500
DTS	0.123
Contribution to DTS	0.0176

2 Treasury and Spread Decomposition

For small changes,

$$\frac{\Delta P}{P} \approx -D_T \Delta y_T - D_s \Delta s.$$

Treasury duration measures exposure to the benchmark curve. Spread duration measures exposure to the bond's compensation over that curve. DTS rewrites spread risk for a proportional spread shock:

$$\text{DTS} = D_s s, \quad \frac{\Delta P}{P} \approx -\text{DTS} \frac{\Delta s}{s}.$$

Portfolio value (\$)	3.5000e+07
Treasury DV01 (\$/bp)	1.8020e+04
Spread DV01 (\$/bp)	1.6440e+04
Portfolio DTS	7.5600e-02

dtype: float64

Equal market values do not imply equal credit risk. Wider-spread bonds can contribute disproportionately to DTS even when their spread duration is shorter.

3 Carry Versus Widening Breakeven

i Audit the utility position

The utility's \$12 million market value and spread duration of 4 give $\$12,000,000 \times 4 \times 0.0001 = \text{\$4,800 per bp}$ of spread risk. A 25 bp widening therefore costs approximately \$120,000. Its 78 bp annual spread carry is only \$93,600. Even a small-looking spread move can erase a year's spread income; spread is quoted per year, but a price loss can occur today.

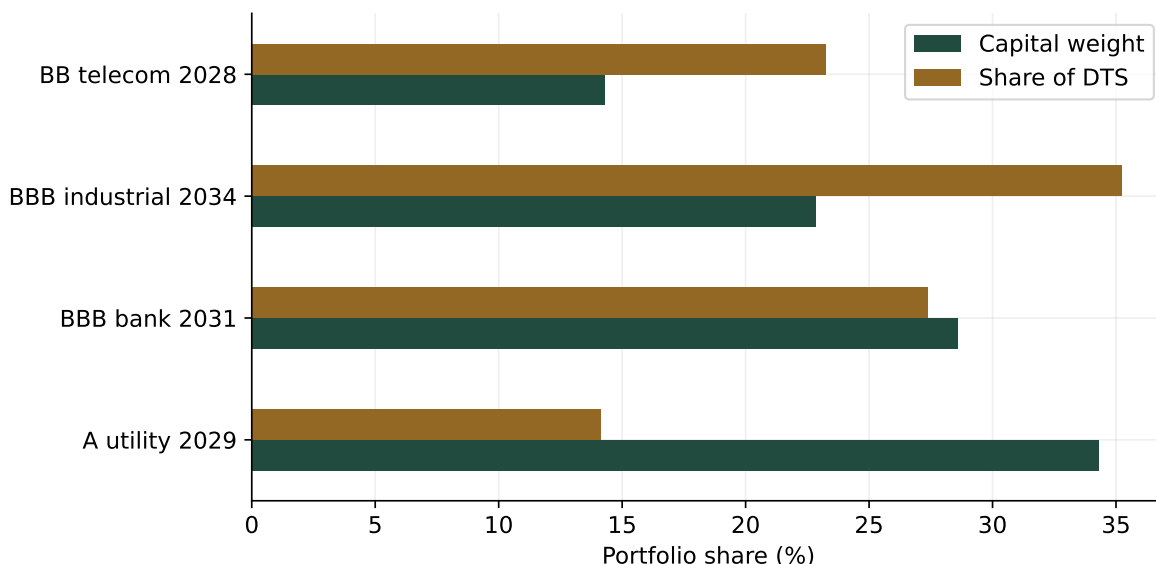


Figure 1: Market-value weights versus shares of DTS risk; all inputs are illustrative.

Approximate annual spread carry is $MV \times s$. The spread widening that consumes one year of spread carry is

$$\Delta s_{BE} \approx \frac{s}{D_s}.$$

This is a mark-to-market breakeven, not a total-return forecast. It ignores rolldown, Treasury carry, default, taxes, and financing.

Bond	Annual spread carry (\$)	Widening breakeven (bp)
A utility 2029	93,600	19.5
BBB bank 2031	142,000	27.8
BBB industrial 2034	148,000	29.4
BB telecom 2028	205,000	136.7

4 Expected Loss and Liquidity Reserve

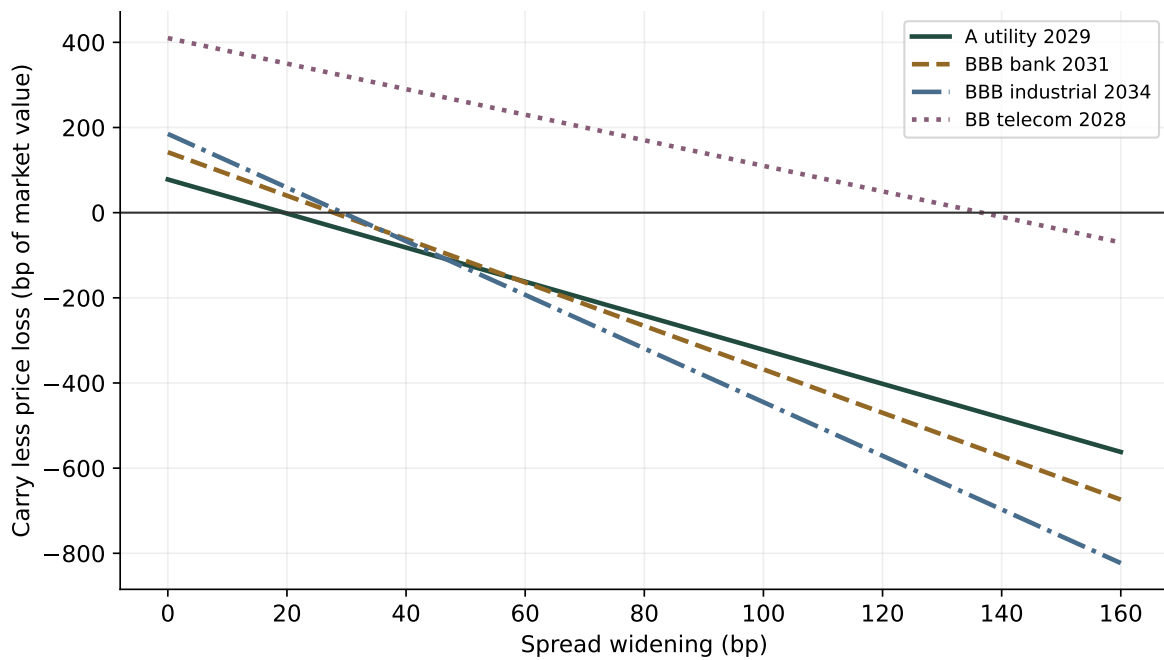


Figure 2: Each crossing is the widening that consumes one year of spread carry, before default and liquidity allowances.

Expected credit loss is

$$EL = PD \times LGD \times EAD, \quad LGD = 1 - \text{Recovery Rate}.$$

Spread is not expected loss. It may also compensate for systematic credit risk, uncertainty, liquidity, taxes, and supply-demand effects.

	Annual spread carry (\$)	Expected loss (\$)	Liquidity reserve (\$)
Bond			
A utility 2029	93,600	10,800	9,600
BBB bank 2031	142,000	24,000	15,000
BBB industrial 2034	148,000	36,400	17,600
BB telecom 2028	205,000	87,500	22,500
Carry after EL and liquidity (\$)			
Bond			
A utility 2029	73,200		
BBB bank 2031	103,000		
BBB industrial 2034	94,000		
BB telecom 2028	95,000		

5 Mild and Severe Scenarios

The scenarios distinguish a benchmark-rate shock from issuer spread shocks. Expected loss is shown as a separate one-year planning measure rather than added to scenario P&L.

	Treasury P&L (\$)	Spread P&L (\$)	Total mark-to-market P&L (\$)
Scenario			
Mild slowdown	360,400	-546,600	-186,200
Severe recession	1,351,500	-2,156,400	-804,900

6 Why the Risks Differ

- **Rate risk** is exposure to the benchmark Treasury curve and can often be hedged with Treasury futures or swaps.

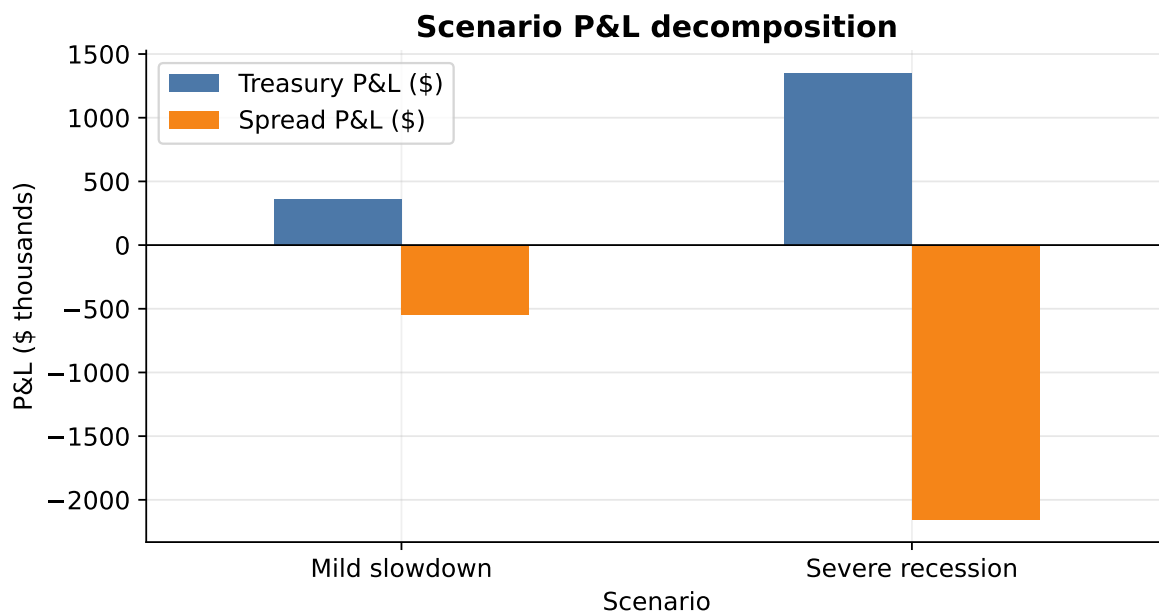


Figure 3: Falling Treasury yields can partly offset, but do not eliminate, spread losses

- **Spread risk** is mark-to-market exposure to changing required compensation for credit and liquidity risk.
- **Default risk** is a discrete credit event with uncertain recovery; PD and LGD describe expected loss, not daily spread volatility.
- **Liquidity risk** is the cost or inability to trade near an observed price, often worst when spreads are already widening.

7 Validation Checks

DTS is positive	True
Expected loss is nonnegative	True
Severe spread loss exceeds mild spread loss	True
Portfolio DTS reconciles to weighted bond DTS	True
dtype: bool	

8 Financial Interpretation

A wide spread is not automatically attractive. The analyst should ask how much of the spread may be consumed by expected loss and liquidity, how much widening erases carry, and

whether falling Treasury yields are likely to offset spread losses in stress. DTS is useful for proportional spread shocks; DV01 is useful for basis-point shocks; neither replaces explicit default and liquidity analysis.

9 Guided decision and market handoff

1. **Hold Treasury yields fixed.** Compare a 25 bp widening in every bond with a 25% increase in each bond's spread. DV01 ranks the first exposure; DTS addresses the second. The shocks are not interchangeable.
2. **Allow a Treasury rally.** Use the severe scenario to explain why government yields falling does not guarantee a gain on a corporate bond.
3. **Write the recommendation.** Cite carry breakeven, spread-loss concentration, and the assumption about default. Do not add expected default loss to the mark-to-market stress as though they were independent realized losses.

Inspect [FINRA's security and trade lookup](#) for an issuer's terms and trade history. Record the benchmark, timestamp, callable status, and spread definition. A yield difference, a Z-spread, and an OAS are not interchangeable inputs. Keep the embedded four-bond case if a comparable security or permitted dataset is unavailable; the supplied PDs are assumptions, not ratings-agency estimates.

10 Interactive explorer

Apply the same proportional spread change to all four bonds while keeping Treasury yields fixed. At +25%, the 78 bp utility spread widens by 19.5 bp, while the 410 bp telecom spread widens by 102.5 bp. These are immediate price changes, with no annual carry or default loss added.

The HTML control selects a precomputed scenario. In the PDF, use the preceding figures and tables and rerun the displayed notebook inputs to explore other values.

11 Mini Project

Project brief. Compare a uniform +25 bp widening with a 25% proportional widening in each position's spread. Report bond-level and total losses for both shocks. Rank each bond's DTS contribution, then compute the carry cushion after expected loss and the supplied liquidity reserve. Recommend one exposure to review, distinguishing physical expected loss from market-implied compensation.

Deliverable: a one-page decision memo with a conventions statement, reproducible calculations, a scenario table, and a limitation that could change the decision.

Worked solution: calculations and decision memo

	25 bp loss (\$)	25% widening loss (\$)	DTS contribution
Bond			
A utility 2029	120,000	93,600	0.0107
BBB bank 2031	127,500	181,050	0.0207
BBB industrial 2034	126,000	233,100	0.0266
BB telecom 2028	37,500	153,750	0.0176

	Net annual carry proxy (\$)
Bond	
A utility 2029	73,200
BBB bank 2031	103,000
BBB industrial 2034	94,000
BB telecom 2028	95,000

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25 bp loss ($)          411000.0000
25% widening loss ($)   661500.0000
DTS contribution         0.0756
Net annual carry proxy ($) 365200.0000
dtype: float64

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A fixed basis-point move is measured by spread DV01; a proportional move is measured by DTS. Wider spreads create a larger absolute shock in the proportional scenario. Review the largest DTS contributor alongside its capital weight and net carry rather than ranking by yield alone. The carry calculation is a rough annual compensation proxy, not a predicted total return: spread repricing, defaults, and liquidity are distinct risks. Physical default estimates need not equal risk-neutral default probabilities implied by market prices.